

IT430 E-Commerce

`Short Notes`

Lesson `19 to 45`

``With Key Points``

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Lesson19

MAINTAINING STATE IN A STATELESS SYSTEM

Stateless HTTP requires scripts or back-end development to maintain state in ecommerce transactions. To link different clicks/steps, programmers can use options like Registration information, item selection, and Confirm Order. These methods help link and maintain state among these steps.

1. Temporary File on Server:

- Create a temporary file on the server side.
- Code scripts behind each step (e.g., Register, Add to Cart, Confirm Order) to store information in the temporary file using a common ID.
- At the end, the Confirm Order script can retrieve the information from the temporary file and store it in a database using an insert query.

2. Client-side Cookies:

- Use cookies to store information regarding Registration and Add to Cart steps on the client side.
- When the user reaches the Confirm Order step, the cookies containing relevant information can be sent back to the server and stored in a database against a common ID.

3. Hidden Form Fields:

- Utilize hidden form fields within HTML forms to maintain state.
- For example, in the Add to Cart form, hidden fields can be used to store item names and codes/values.
- When the user selects items and presses Add to Cart, a script can open a new page with a "Confirm Order" button.
- In the new form, the item codes/values of the selected items can be present but hidden.
- When the user presses the "Confirm Order" button, a script can store the information of selected items in the database against the item codes/values present in the hidden form fields.

Example - Hidden Fields

```
<FORM NAME="Form1">  
  <INPUT TYPE="HIDDEN" NAME="Shirt1" VALUE="25">  
  <INPUT TYPE="CHECKBOX" NAME="Check1">Blue  
  Cotton Shirt<br>
```

```

<INPUT TYPE="HIDDEN" NAME="Shirt2" VALUE="26">
<INPUT TYPE="CHECKBOX" NAME="Check2">Green
Cotton Shirt<br>
<INPUT TYPE="HIDDEN" NAME="Shirt3" VALUE="27">
<INPUT TYPE="CHECKBOX" NAME="Check3">White
Silk Shirt....
<P>
<INPUT TYPE="submit" NAME="Go" VALUE="Register!">
<INPUT TYPE="RESET" VALUE="Reset!">
</FORM>

```

Another option is to keep everything in databases

- Store information from each step directly in appropriate database tables.
- Link the information using a common ID.
- This option requires more resources and time compared to other methods.

The main purpose here is to provide you with a broad concept of how state can be maintained in a stateless system.

Client server architecture

Client-server architecture refers to a model where the processing and distribution of tasks are divided between client devices and server systems. It can be categorized into different tiers based on where the processing of data occurs.

Key points: #Imp

- **1-Tier Architecture:** All processing happens in one location. For example, a mainframe with connected display terminals or a file server that only stores files.
- **2-Tier Architecture:** Processing is split between the client and server. For instance, clients handle HTML code and web page requests, while servers process those requests.
- **3-Tier Architecture (different processing zone):** Additional tiers are introduced, such as a separate tier for the database management system or application software, improving scalability and resource management.
- **N-Tier Architecture:** The client-server model can have more tiers based on specific needs. For example, 4-tier architecture may include a separate tier for payment processing.

Operating systems for web server #Imp

A web server is a machine that serves client requests on the internet, combining **hardware and software**. The choice of server depends on the volume and type of web activities. **Common operating systems** include Microsoft Windows NT Server, Windows 2000 Advanced Server, Microsoft.NET Server, Linux, and Unix-based systems like Solaris.

Open source software, such as Apache HTTP server, MS Internet Information Server (IIS), and Sun ONE web server, is popular and secure. **Web server hardware**, such as hard disk drives and processors, is typically more memory-intensive than desktop PCs. Companies like Dell and Hewlett Packard offer configuration tools for users to design their own web servers.

Performance of web sever

- Performance of web servers is assessed based on throughput (number of requests handled) and response time (time taken to process a request).
- Throughput indicates the server's capacity, while response time measures its efficiency.

Web Server Hardware Architectures:

1. E-commerce sites can adopt multi-tier architectures with multiple computers in each tier.
2. Centralized architecture uses fewer powerful computers per tier, while decentralized architecture distributes workload among many less powerful computers.

Web Hosting Choices:

- ISPs offer web hosting options such as shared hosting, where multiple websites are hosted on the same server.
- Dedicated hosting provides a dedicated server for exclusive use.
- Co-location hosting allows clients to install and maintain their own server hardware in the ISP's facility.
- Self-hosting involves running a server in-house.

E-commerce software

E-commerce software varies in size and objectives, with various software and hardware products developed for building sites. It typically includes **catalog display, shopping cart capabilities, transaction processing.**

Large E-commerce sites utilize software for additional features and capabilities.

- Middleware
- Application integration
- Web services
- ERP integration
- Supply chain management
- CRM, content management
- Knowledge Management Software

- Catalog display:

A static catalog is a basic HTML list that needs manual editing. A dynamic catalog uses a database to store information about items, allowing for photos, detailed descriptions, and search functionality.

- Shopping cart:

In the past, users had to remember and type in information for form-based shopping carts. Nowadays, electronic shopping carts keep track of selected items, display their details, and allow

adding/removing items with a simple click. All item information is automatically stored in the cart.

- Transaction processing:

When a customer clicks the checkout or confirm order button, transaction processing takes place. This involves payment processing, including calculations for taxes and shipping costs. Some software can connect to shipping companies' websites to get updated shipping rates.

Advanced functions of e-commerce software

- Middleware:

Large companies use middleware software to connect their e-commerce and accounting systems, enabling data synchronization.

- Application integration:

E-commerce software should be compatible and integrated with the company's existing database management software. For

example, if there's an inventory database, the e-commerce software should support it.

- **Web services:**

Web services are software tools that allow different programs to communicate over a network using standard protocols. They can gather financial information from multiple sources, obtain price and delivery details from vendors, and facilitate order placement and tracking.

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INTEGRATION WITH ERP SYSTEMS

Enterprise Resource Planning (ERP) #Imp

Enterprise Resource Planning (ERP) is a concept that integrates all aspects of a business e.g., accounting, logistics, manufacturing, marketing, planning, project management etc. at a single place. An ERP system such as SAP is expensive. E-commerce sites/software has to fully integrate with ERP software, wherever it is used.

Customer Relationship Management Software (CRM) #Imp

Software that helps businesses understand what customers want and tailors products or services to meet their needs by collecting data from customer interactions on a website.

Supply Chain Management (SCM) Software:

Software that helps companies manage the flow of goods from start to finish, including planning and coordination with partners, such as forecasting demand and managing warehouses and transportation.

Content Management Software:

Software that helps companies organizes and control large amounts of data, images, and files used for business, making it easier to access and use.

Knowledge Management Software:

Software that helps companies manage and extract useful information and insights from various sources like documents and emails, making knowledge sharing and collaboration more effective.

E-commerce Software

Following are the names of some well-known e-commerce software:

- Intershop Enfinity
- IBM's WebSphere Commerce Professional Edition
- Microsoft Commerce Server 2002.

Agents:

Autonomous software programs that perform tasks like comparisons, filtering, and negotiations. They can be used for various purposes such as web crawling, auctions, and health monitoring. For example, buyer and seller agents can negotiate deals on behalf of users.

Security issues over the internet:

Security concerns are a major obstacle to the growth of e-commerce. When sensitive information like credit card numbers is entered online, it becomes vulnerable to potential misuse by millions of people on the internet.

Back doors and Trojan horses: #Imp

Back doors are hostile programs that provide attackers with remote access to compromised machines. Trojan horses are seemingly harmless programs that actually have malicious intent. They are named after the historical story of the deceptive wooden horse used in war.

Viruses and worms:

Viruses and worms are malicious programs that can spread between computers through email attachments or over networks. Worms can travel independently and replicate themselves, while viruses require human action to activate and can modify existing programs. The 'Love Bug' worm, triggered by a student in the Philippines, caused significant damage by spreading through email attachments and compromising millions of computers worldwide.

Hackers:

Individuals who use their programming skills to gain unauthorized access to computers and networks.

Active contents, Active X controls: Active content refers to programs embedded in web pages that can perform actions. Malicious active content, such as Applets, JavaScript, and Active X controls, can be used by hackers to install hidden services, collect sensitive information, or cause damage to the user's system.

Outside attacks on a network:

Eavesdropping/sniffing/snooping:

Hackers use network-monitoring software to monitor network traffic and gather sensitive information, such as keystrokes, from users.

Password attacks:

Hackers gain access to a network by eavesdropping and obtaining user account IDs and passwords, allowing them to modify, delete, or reroute network data.

IP address spoofing:

Hackers manipulate data packets by changing the source IP address, redirecting the information to their own machines instead of the intended destination.

Man-in-the-middle attacks: Attackers intercept and manipulate data between sending and receiving machines, either by diverting packets to their own machines or substituting the IP address of a genuine website with their own.

Denial of service (DOS) attacks: Attackers overwhelm network services or applications by sending invalid data, causing them to become overloaded or shut down. This can be used to take down websites by flooding them with requests, making them inaccessible to genuine users. Distributed denial of service (DDOS) attacks involves using compromised systems as sources for further attacks.

Virus protection

1. Install anti-virus software (e.g., McAfee, Norton) to detect and remove viruses.
2. Avoid downloading plug-ins from untrusted sources. Only download them from official vendor websites.
3. Scan newly obtained disks, programs, and files for viruses before using them.
4. Consider installing a firewall to block unauthorized access and reduce the risk of virus attacks.

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FIREWALLS

A firewall is a hardware and software system protecting an organization's internal network from external attacks. It examines and filters data based on rules, with three main types.

Packet Filter Firewall:

It operates at the network and data link layers of the OSI model. It uses rules to examine incoming and outgoing packets based on information like IP addresses and port numbers to determine whether to allow or block them.

2. Circuit Level Firewall:

Similar to a packet filter firewall, it operates at the transport layer. It can monitor TCP/IP connections and verify their proper establishment, providing protection against hijacked connections. It can also hide internal network IP addresses through Network Address Translation (NAT).

3. Application Gateway Firewall:

It operates at the application layer. It filters traffic based on application-layer protocols like FTP, HTTP, or SMTP. It enforces user authentication and can restrict outgoing requests, such as blocking access to certain websites or preventing downloads of potentially harmful programs. Hybrid firewalls combine circuit level and application gateway features for enhanced security.

Proxy server

A proxy server acts as an intermediary between a trusted internal network and the untrusted internet. It provides several functions to enhance network security and optimize web browsing:

Mainly, it can do three things

IP Address Masking:

When a user sends an HTTP request, the proxy server can replace the user's IP address with its own. This helps to hide the identity and location of the requesting machine, providing an additional layer of anonymity and security.

Web Page Caching:

The proxy server can store frequently accessed web pages in its cache memory. This allows it to quickly respond to subsequent requests for the same web pages without needing to fetch them from the original web server. Caching reduces the load on the network and speeds up the delivery of web content to users.

Filtering and Firewall Capabilities:

Proxy servers can be configured to filter and block certain web pages or content based on predefined rules. This enables the enforcement of access controls and content restrictions, enhancing security by preventing users from accessing malicious or unauthorized websites.

Virtual private network (VPN)

A virtual private network (VPN) is a cost-saving method for remote access to a company's head office network. It allows clients to connect to a server machine in the head office network through a direct telephone line, using the Point to Point Protocol (PPP) to exchange data packets. VPN connections are managed by PPTP, ensuring secure communication over the public internet. Tunnel clients incur only the cost of a local call to the ISP, while dial-up connections require long distance calls. VPNs can also form extranets, allowing manufacturing companies to connect to suppliers and distributors.

Security – the biggest challenge

The issue of computer and data security is the biggest challenge in e-commerce growth, as it can damage databases, terminate server software, or alter information. International organizations like the Computer Emergency Response Team (CERT) and Systems Administrator, Audit, Network and Security Institute (SANS Institute) have been formed to combat security threats to computers and networks. Experts have proposed cryptography as the best response to tackle this security issue.

Cryptography#Imp

Cryptography is a technique that converts messages into unintelligible or non-understandable forms, ensuring that even unauthorized individuals cannot understand them. It has thousands of years of history and has been used for military purposes, including protecting religious secrets and securing scientific and industrial secrets. Cryptography has also been used for non-military purposes, such as protecting religious secrets and ensuring the rapid commercialization of the internet. Without cryptography, businesses, individuals, and individuals would not be able to conduct online transactions.

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CRYPTOGRAPHY

Cryptography is a set of mathematical techniques used to ensure information confidentiality. Encryption involves scrambling a message with a key, while decryption involves unscrambling it using an appropriate key. Keys are randomly generated using cryptographic algorithms. PGP is a popular cryptographic system for public use.

There are two types of cryptography:

Symmetric and Asymmetric: #Imp

Symmetric cryptography uses the same keys for encryption and decryption, **Asymmetric cryptography** uses a pair of public and private keys for encryption and decryption.

Digital certificates

Digital certificates are electronic documents that validate the ownership of a public key and contain certification information. They are created following the X.509 standard format and help establish trust and security in digital communications.

Certification authority (CA)

Certification authorities (CAs) are trusted organizations that verify the association between a person and their public key. They issue digital certificates containing the person's public key, certification information, and the CA's electronic signature. CAs work in a hierarchical structure, with a root CA at the top issuing certificates to lower-level CAs, creating a chain of trust. When sending an encrypted message, if the recipient's public key is certified by a reputable CA, it provides assurance of authenticity. This system helps establish trust and ensures the authenticity of individuals in internet communication.

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HASH FUNCTION AND MESSAGE DIGEST

#Imp

- Hash function is a one-way mathematical function applied to a message, resulting in a unique message digest.
- Message digest is a large number (typically 128 to 256 bits) that represents the message.
- Each message has a unique message digest, making it practically impossible to reverse-engineer the original message.
- Hash functions ensure message integrity by confirming that a message hasn't been altered.
- An example of a hash function is XORhash, which performs an XOR operation on the characters of a message.

To encrypt and digitally sign a message:

1. The sender applies a hash function to the message, obtaining a message digest.
2. The message is encrypted using the receiver's public key, making it confidential.
3. The sender adds their private key to the message digest, creating a digital signature.

4. The receiver decrypts the digital signature using the sender's public key, revealing the message digest.
5. The receiver uses their private key to decrypt the message and applies the same hash function to generate a new message digest.
6. By comparing the received message digest with the generated one, the receiver verifies the message's integrity.

Public Key Infrastructure (PKI)

PKI is a structured system for managing digital certificates and storing them, involving a certification authority. It is used in online contracts, e-banking and electronic payment systems.

Key Length

A cryptographic key is a string of binary digits, with 1 bit representing two possible keys, 2 bits representing four possible values, and 3 bits representing eight possible values. The number of keys can be calculated using the formula: Number of keys = $2^{(\text{number of bits})}$.

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SYMMETRIC KEY ALGORITHMS

In symmetric key cryptography, cryptographic algorithms are evaluated based on their key length. Here are some popular symmetric key algorithms and their key lengths:

1. DES (Data Encryption Standard) - 56 bits
2. IDEA (International Data Encryption Algorithm) - 128 bits
3. RC2 (block cipher) - 1 to 2048 bits
4. RC4 (stream cipher) - 1 to 2048 bits
5. Rijndael - 128 to 256 bits

Attacks on Symmetric Key Algorithms

1. Key Search (Brute Force) Attacks:

(Key search attack is like trying every possible key until the right one is found)

In this type of attack, the attacker tries to decrypt the message by attempting every possible key. The longer the key length, the more difficult it becomes to determine the correct key.

2. Cryptanalysis

(Cryptanalysis involves using smart techniques to break the encryption without knowing the key)

Cryptanalysis involves using advanced mathematics and computing power to break encryption algorithms without knowing the key. This attack aims to find weaknesses or vulnerabilities in the algorithm itself.

3. System-Based Attacks:

(System-based attacks focus on exploiting weaknesses in the overall system rather than directly attacking the algorithm)

System-based attacks target the cryptographic system that uses the algorithm, rather than attacking the algorithm directly. These attacks exploit vulnerabilities in the implementation, key management, or other components of the system.

Public Key Algorithms

Following is the list some popular public key algorithms:

- **DSS** – Digital Signature Standard based on **DSA** (Digital Standard Algorithm) – key length is between 512-1024 bits
- **RSA**
- **Elliptic Curves**

Attacks on Public Key Algorithms:

Key Search Attacks: #Imp

These attacks try to figure out the private key by working backward from the public key. They use mathematical methods to break the encryption and access the private key. Stronger and longer keys make it harder for attackers to succeed.

Analytical Attacks:

These attacks exploit weaknesses in the mathematical foundations of the encryption system. By finding flaws in the algorithms, attackers can bypass the encryption and gain unauthorized access to the information.

Quantum Computing Attacks

Quantum computers are super-powerful machines that can potentially break certain types of encryption used in public key algorithms. This poses a challenge because existing encryption methods may not be secure against future quantum computers. New encryption algorithms are being developed to resist these attacks.

Electronic Payment Systems:

Electronic payment systems, including credit cards, electronic checks, electronic cash, and micro-payment systems like military and payword, utilize cryptography to ensure confidentiality and security of payment information on the internet.

The Process of Using Credit Cards

1. Apply for a credit card from a bank and receive the physical card after approval.
2. Present the credit card to the merchant for payment.
3. Merchant requests authorization from the card brand and completes the transaction.
4. Merchant sends transaction details to their bank and pays a fee.
5. Acquirer bank requests payment from the card brand.
6. Card brand asks the issuing bank to transfer funds from the cardholder's account to cover the transaction.

Credit cards allow users to make purchases and pay later, with interest charged by the issuing bank. Debit cards, on the other hand, use funds directly from the user's bank account.

Virtual PIN Payment System#Imp

The Virtual PIN Payment System was an early online payment system introduced in 1994. It allowed buyers and merchants to make transactions without sharing credit card details directly. Buyers registered with First Virtual and received a pass phrase called the Virtual PIN. Merchants also registered and received a merchant Virtual PIN. When a buyer made a purchase, they entered their Virtual PIN on the merchant's website. The merchant verified the PIN with First Virtual, sent the goods, and received payment through an electronic clearing system. Buyers confirmed their satisfaction with the goods through email. If fraud was reported, the Virtual PIN was blacklisted to prevent unauthorized use.

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VIRTUAL PIN PAYMENT SYSTEM

A virtual PIN can be compromised through eavesdropping, allowing bogus purchases before being blacklisted. Stolen credit card numbers can also be used to set up virtual PINs with controlled email addresses. Buyers' credit card accounts are billed for accumulated charges, while merchants are credited accordingly. FV accounts for both parties, taking a commission per transaction.

Advantages of the Virtual PIN Payment System: #Imp points

- 1. Simplicity:** No special software is required at the front end, making it easy for users to make payments.
- 2. No encryption:** The absence of encryption simplifies the payment process and eliminates the need for complex encryption mechanisms.
- 3. Suitable for low-cost information items:** It is ideal for purchasing online articles, journals, music, and other low-cost digital goods.
- 4. Limited financial loss for merchants:** In case of fraud, the merchant typically incurs only lost sales or the creation of extra copies of digital items.

Disadvantages of the Virtual PIN Payment System:

- 1. Mandatory pre-registration:** Both buyers and merchants are required to register with First Virtual (FV) before using the system, adding an additional step to the setup process.
- 2. Dependence on bank accounts and credit cards:** Merchants need to maintain a bank account, while buyers must possess a credit card, which can be a barrier for some users.
- 3. Limited security for merchants:** The merchant delivers goods before receiving payment, exposing them to a higher risk of financial loss compared to other payment systems.
- 4. Decline in popularity:** The Virtual PIN system's popularity decreased after 1998 due to the introduction and advancement of encryption-based payment mechanisms, which offered enhanced security features.

Centralized Account Payment Model #Imp

The Centralized Account Payment Model is a widely used online payment system, used by platforms like PayPal, E-gold, and Amazon.com Payments. Both buyers and merchants have accounts with a centralized online financial institution. Users can fund accounts using credit/debit cards or prepaid cards. Payments are authenticated using an account identifier and password, and communication between users and banks is protected using SSL encryption. Payees are notified of payments via email and can confirm them through SSL.

Some payment systems may send verification questions to the payee, allowing for transfers if the answer is accepted as proof of identity. Transaction fees may be charged by the payer, payee, or both. All participants must have accounts with the same central payment system or bank. Payees can withdraw money from their accounts through the Automated Clearing House (ACH) system.

OR

The Centralized Account Payment Model

- Used by platforms like PayPal, E-gold, and Amazon.com Payments.
- Involves accounts with a central online financial institution for both buyers and merchants.
- Authentication using an account identifier and password, protected with SSL encryption.
- Payees notified of payments via email and can withdraw money through ACH.
- Transaction fees may apply.

Electronic Checks

The Financial Services Technology Consortium (FSTC) is a group of US banks, research agencies, and government organizations that introduced electronic checks in 1993. These checks contain instructions for a bank to make a specified payment to a payee using symmetric and asymmetric cryptography. Participants use an electronic checkbook device, which combines secure hardware and software. The payer

generates a blank check, digitally signs it, and the bank attaches a certificate. The encrypted check is sent to the payee via email, where they decrypt the envelope using their private key, decrypt the information, counter-sign the check, and forward it to their bank.

The bank clears the check using traditional ACH methods, debiting the payer's account and crediting the payee's account. EFT refers to electronic funds transfer.

OR

The Financial Services Technology Consortium (FSTC) introduced electronic checks in 1993. These checks use symmetric and asymmetric cryptography and are sent via email. Payers use a digital signature, and payees decrypt the check with their private key and counter-sign it before the banks clear the payment through traditional ACH methods. EFT facilitates the electronic funds transfer.

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E-CASH PAYMENT SYSTEM

DigiCash developed e-cash, enabling anonymous buyers and involving three participants: buyers, merchants, and a bank. The system uses symmetric and asymmetric cryptography for security. Buyers withdraw coins, store them in a cyber wallet, and merchants send them to the bank to verify their authenticity and credit the merchant's account.

E-cash Coins

E-cash coins are a payment system currency consisting of three elements: serial number, key version, and serial no. Each coin has a unique value, partly minted by the client and partly by the e-cash bank. For example, a one-dollar coin would have the following formula: Coin = Serial#, keyversion, {Serial #}SK bank's \$1 key.

Minting of the coin#Imp

A client's Cyber wallet generates a random long serial number for coin minting, which is blinded and sent to an e-cash bank for signatures. The bank cannot see the serial number it is signing. A key version, the corresponding public key, is also included in the coin. E-cash banks can have signatures of various denominations, such as 1 dollar, 5 dollars, or 10 dollars. For example, if the client wants to mint a 2 dollar coin, the bank uses its private or secret key of 2 dollars to sign the serial number.

How bank signs blindly? #Imp

Cyber wallet generates a serial number and a blinding factor 'r' for bank signatures, which is only known to the client. The bank uses this blinding factor to sign with its secret key of 2 dollars (d2), which cancels out the product of e2 and d2. The bank cannot see the serial number it is signing, as it does not know 'r'. The client divides the expression with 'r' to obtain the third part of a valid 2 dollar coin, completing the minting process. E-cash banks sign serial numbers and debit the client's account, but clients must maintain their account with the bank. For example, if a client has \$50 in their account and requests 10 coins, the remaining balance after the bank's signatures on the serial numbers would be \$40.

A client wallet or cyber wallet can generate serial numbers, store coins, and make and accept payments. It can withdraw coins from the e-cash bank and deposit them at the payment stage. E-cash banks sign serial numbers, maintain client and merchant accounts, and maintain a

database of coins sent for payment. Clients make payments to merchants, who store minted e-cash coins in the wallet. E-cash banks decrypt these coins using their private key and compare them with their spent coins database to verify their validity.

If a revealed serial number is not present, the bank sends a valid indication to the merchant, adding the serial number to prevent future consumption. The merchant then sends the goods and payment receipt to the client.

OR

In the banks blind signing process for e-cash

- Client generates a serial number and blinding factor 'r'.
- The bank signs with its secret key using 'r' to blind the serial number.
- The bank cannot see the serial number it signs.
- Client completes minting process by dividing the signed expression with 'r'.
- E-cash coins are withdrawn from the bank, reducing the client's account balance.
- E-cash banks also maintain merchant accounts and verify coin validity during payments.
- If a serial number is new, the bank adds it to prevent future use.

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E-CASH PAYMENT SYSTEM

How anonymity is ensured in e-cash payment system? #Imp

Anonymity (گمنامی) in e-cash payment systems ensures that the identity of the client/buyer is not disclosed. The system has two main stages: minting and deposit. Minting involves signing the serial number to provide a valid e-cash coin, while depositing coins for verification. The bank knows the serial number but has no clue about the specific client who sent them for payment purposes. This breaks the relationship between the client and the serial number at the minting and deposit stages to ensure anonymity.

The concept can be illustrated as follows:

Minting Stage:

Serial number unknown; Client (known);

Deposit Stage

Serial no. known; Client (unknown)

Withdrawing Coins

A single request to a bank allows for the withdrawal of multiple coins of different denominations. The client's private key is used to sign the request, which is encrypted with a symmetric key and public key of the bank. The bank then mints the coins and forwards them to the client/buyer.

E-cash Purchase

Merchant sends payment request with Payreq format,
`Payreq={currency,amount,timestamp,merchant_bank ID, merchant_accID, order description}`

Which is automatically assembled by cyber wallet and paid to client, ensuring accurate payment

Making the Payment

Payments are encrypted with a bank's public key, preventing merchant access. Payment information is forwarded to the bank with encrypted coins during deposit, with only the order description's hash included, preventing the bank from knowing the order details.

Proving the Payment

Client generates a payment code, which is included in payment information as a hash for later verification.

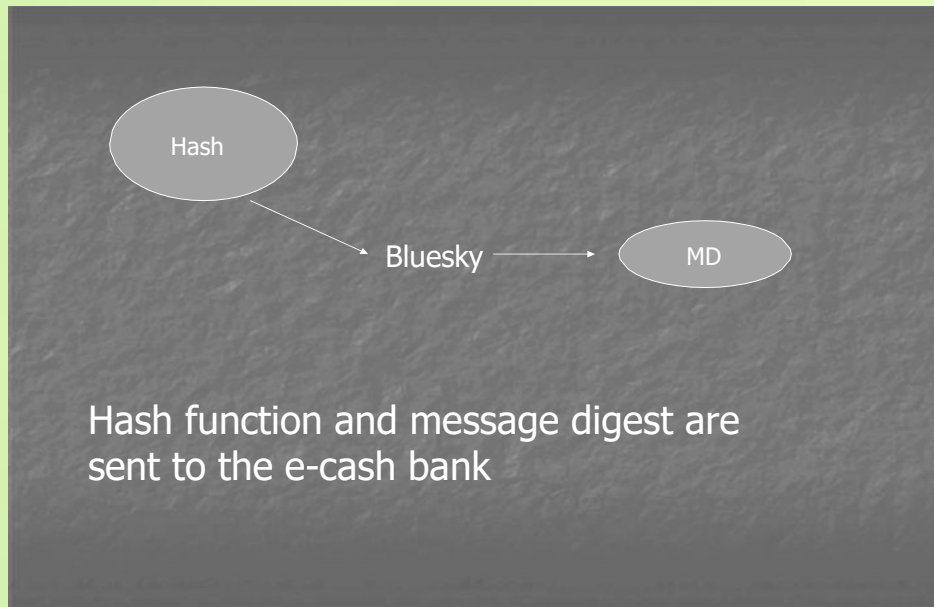


Fig. 1 #Page119

In this system, the client chooses a secret code like "Bluesky" and applies a hash function to it, creating a unique message digest. The hash function and digest are sent to the bank. If a dispute arises, the client can provide the original code to the bank and request them to apply the hash function. If the resulting digest matches the one stored by the bank, it proves the client made the payment since only they should know the original code.

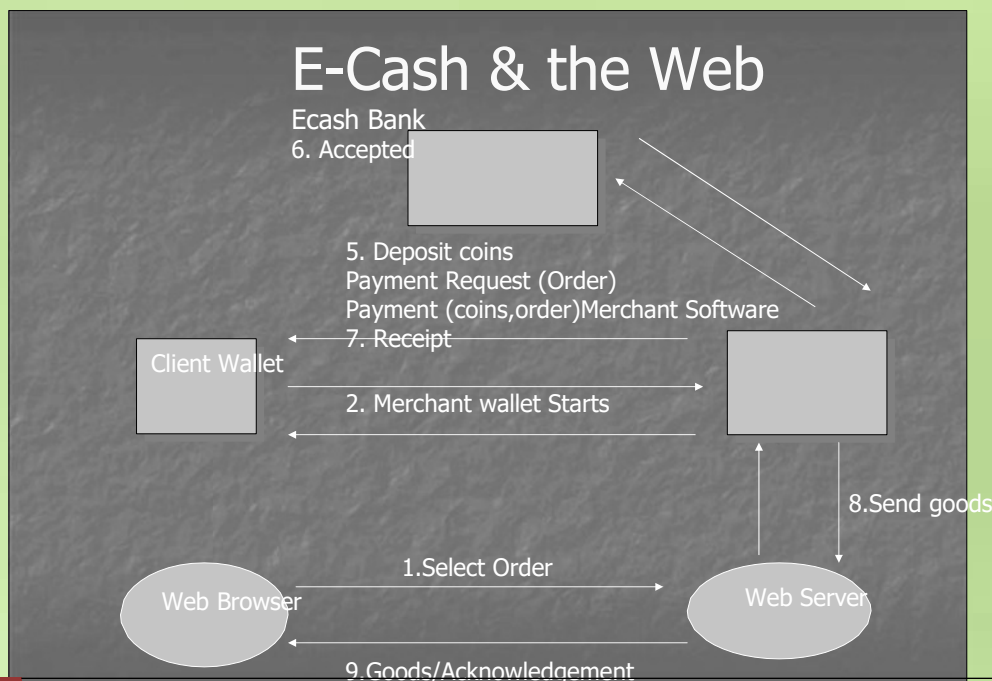
Payment Deposit

Payment deposits are encrypted with a bank's public key and encrypted. E-cash banks check the validity of spent coins to prevent double spending. If valid, the bank credits the merchant's account. If the client sends valid coins worth \$10, the merchant receives \$100. The

merchant can request the bank to transfer the amount to their acquirer bank via ACH, and the bank charges a commission for its services. E-cash banks play a backbone role in this system, charging a commission depending on their policy.

Lost Coins

In case of network failure or computer crashes during payment transactions, lost coins may occur. Bank sends signed blinded coins from the last 16 withdrawals to the client, who uses a wallet-specific blinding factor to reveal the serial number. The bank verifies the coins' spent status and credits the client's account with the unspent coins value.



The client machine has a client wallet and web browser, while the server and merchant software are installed on the server. A client selects an order, and the merchant software initiates a payment request. The client wallet pays through e-cash coins and the merchant deposits the coins to check their validity. If the coins are valid, the merchant sends an acceptance message, and the web server sends goods, acknowledging the transaction to the client's browser.

Lesson 28

SECURE SOCKET LAYER (SSL)

#Imp

SSL is a Netscape Communications-developed protocol used in many browsers at the TCP/IP layer of the OSI model. It uses symmetric and asymmetric cryptography. The word "https" in a URL indicates the web server's SSL enabled status. To configure a client machine for SSL, follow these steps:

1. Open Internet Explorer and navigate to Tools menu.
2. Select Advanced tab.
3. Check the "use SSL" option.

SSL Handshake#Imp

SSL Handshake is a protocol used by web browsers to establish communication between clients and servers. It uses a cipher suite to select the strongest encryption algorithm and authentication method. The SSL handshake process involves a client sending a message to the server, the server sending its digital certificate, and the client generating a secret symmetric key. The server decrypts the message using its private key, and the session key is used for encryption and decryption. SSL is commonly used in online shopping, where customers enter credit/debit card information on web forms for payment.

The SSL handshake is simple, as it is built into many browsers, and no special encryption software is required. However, it also has a drawback: merchants can store unauthorized access to credit/debit card information in their database.

Or

SSL handshake is a process that establishes secure communication between a client and a server during online transactions. They agree on encryption methods and authenticate each other. Once the handshake is complete, sensitive data like credit card information is encrypted and transmitted securely. However, a drawback is that merchants may store decrypted data, which can be a security risk.

Secure Electronic Transaction (SET) #Imp

Secure Electronic Transaction (SET) was developed in 1997 by Visa, MasterCard, Netscape, and Microsoft to address the issue of merchant information remaining with the cardholder. SET involves four entities: cardholder, merchant, certification authority, and payment gateway. Merchants need special SET software for processing transactions, while customers need digital wallet software for storing certificates and card information.

Dual Signature in SET#Imp

In the SET (Secure Electronic Transaction) system, the dual signature scheme is used to protect privacy. It combines two message digests to create a new digest called the Dual Signature Message Digest (DSMD). This process helps hide the customer's credit card information from merchants and the order information from banks, ensuring privacy for both parties involved in the transaction.

In SET, the customer's order and account information are split. Message digests (MD1 and MD2) are obtained by applying a hash function. They are concatenated to form the Dual Signature Message Digest (DSMD). The buyer sends the encrypted order information with the signed MD1 to the merchant, who decrypts and verifies. If accepted, the merchant signs MD1 and sends it to the acquirer bank with an encrypted acceptance letter. The buyer sends encrypted account information, MD2, and DSMD (signed) to the bank, which decrypts and receives all the relevant information.

Or

1. Customer splits order and account info.
2. Creates two message digests: MD1 for order info, MD2 for account info.
3. Concatenates MD1 and MD2 to create DSMD (Dual Signature Message Digest).
4. Buyer sends MD2 and DSMD (both signed) to the acquirer bank.
5. Merchant receives order info and sends MD1 (signed) to the acquirer bank.
6. Acquirer bank checks if MD1+MD2 hash matches DSMD, ensuring a valid transaction.
7. Customer's credit card info remains hidden from merchants, and order info stays confidential from banks.

SETCo

A company that promotes and implements SET specifications, ensuring vendors comply with originators' requirements. Merchants and customers hold certificates from card brands and banks, respectively, acting as root certification authorities in the certification hierarchy.

SSL vs. SET#Imp

SSL focuses on securing credit card data transmission, while **SET** handles the entire transaction securely using dual signatures.

SSL is a browser-based protocol, while **SET** requires software on both the client and merchant sides.

SET utilizes a certificate hierarchy for authentication.

SET can be complex and unstable in certificate distribution, leading to increased transaction costs.

SET transactions are slower compared to SSL.

SET employs a payment gateway for secure information transmission.

E-Business

An e-business refers to a company/entity with an online presence that engages in selling, trading, or transacting over the web. It operates based on policies, operations, technology, and ideology.

Advantages of e-business over traditional business include

Personalized service, high-quality customer service, no inventory cost, global reach, electronic catalogues for convenient transactions, bulk transactions, and improved supply chain management.

Lesson 29 **E-BUSINESS**

Advantages of E-business

Personalized service

The 4 P's of marketing in traditional offline businesses include product, place, price, and promotion. In e-business, personalization is the 5th P, as internet technology allows for the recording of customer information and buying behavior in databases. This information can be used by e-commerce sites to

understand customer needs, provide personalized services, and analyze their buying behavior.

4 P's = Product, place, price and promotion.

High-quality customer service:

Customers can conveniently provide feedback and register complaints, allowing e-businesses to improve their customer support.

No inventory cost: E-businesses can operate without the need for physical stores or inventory storage, reducing overhead costs.

Worldwide reach:

E-businesses have a global presence, enabling them to reach customers worldwide and operate 24/7.

Electronic catalogues:

Online shops utilize electronic catalogues, offering quick and convenient browsing and selection of products.

Bulk transactions:

E-businesses allow customers to make large-volume purchases in a single visit, without the limitations of physical transportation.

Improved supply chain management:

Online presence of suppliers facilitates quick and efficient ordering and delivery of raw materials, leading to streamlined supply chain management and faster delivery to end customers.

Disadvantages of E-business:

Less security:

The issue of security is a major concern in e-commerce, as the internet is prone to hacking and unauthorized access, making customers hesitant to share sensitive financial information online.

Less privacy:

Online businesses can collect and record private information of customers, potentially breaching their privacy rights. Tools like cookies can track customer behavior, raising concerns about data privacy.

No physical proximity with items purchased:

Certain products require physical examination before purchase, such as clothes or perfumes. E-businesses lack the ability to physically touch or test products, which can be a limitation for customers who prefer physical shopping experiences.

Paper catalogues

Paper catalogues advantages,

- Being easier to create
- Reader-friendly
- More portable than electronic ones

Disadvantages:

- Limited product display
- Difficulty updating information
- Difficulty updating product details

Online or electronic catalogs

Advantages:

- Product information is easily updated
- Integrated with the purchasing process
- Offers search and comparison capabilities
- Provides a broad range of information
- Offers voice and motion pictures
- Cost savings are achieved
- Easy customization is possible.

Disadvantages

- Difficult to develop catalogues
- Large fixed cost if used for small no. of products
- Need for customer skill to deal with computers and browsers

E-Business Models: #Imp

- **Storefront Model:**

Sellers interact directly with buyers through their websites, offering online catalogs, secure payment processing, and order fulfillment. Common in B2C transactions

- **Auction Model:**

Users participate in online auctions as buyers or sellers, setting reserve prices and closing deadlines. Auction sites charge commissions on sales.

- **Online Banking:**

Banks offer internet-based banking services, including electronic funds transfer.

- **Online Trading and Lending:**

Trading stocks and shares online, often facilitated by brokerage houses. Online lending provides loans through internet transactions.

- **Online Recruiting:**

Employers and job seekers connect through online recruiting websites.

- **Online News Services:**

Newspapers and magazines provide 24-hour updates and content online.

- **Online Travel Services:**

Customers can make travel arrangements; find discount fares, book flights, hotels, and rental cars online.

- **Online Entertainment:**

Online businesses sell music albums, movie tickets, and video content for easy downloading.

- **Online Automotive Sites and Energy Online:**

Websites for buying and selling cars, spare parts, and energy products (oil, electricity) online

- **Selling Intellectual Property Online:**

E-businesses deal in the sale of intellectual property rights such as patents, trademarks, and trade names.

- **Online Art Dealers:**

Artwork, including pictures, paintings, and posters, can be purchased online at discounted prices.

- **E-Learning:**

Universities and training institutes offer high-quality distance education and downloadable e-books and reading material.

- **Online Service Providers:**

E-businesses that improve policies, procedures, customer service, and operations for other businesses, offering consultancy and professional services

- **Online Shopping Malls:**

Websites that offer a wide selection of products and services from multiple stores, allowing customers to make purchases from different stores in a single transaction

- **Portal Model:**

Portals provide a wide range of information in one place, either as horizontal portals (covering various topics) or vertical portals (focusing on a specific area of interest). Some portals include online shopping options.

Brick-and-Mortar vs. Click-and-Mortar businesses

#V.Imp

Brick-and-Mortar businesses are traditional businesses with physical stores, relying on face-to-face interactions.

Click-and-Mortar businesses are hybrid businesses with an online presence, allowing customers to shop online or visit a physical store for purchases.

Lesson 30

E-BUSINESS REVENUE MODELS

Experts have identified following revenue models on the web: **#Imp**

1. Web Catalogue Revenue Model:

E-businesses generate revenue by offering electronic catalogs and shopping carts, selling products or services online, and receiving payment from customers.

2. Digital Content Revenue Model:

E-businesses distribute digital content such as information services, news, and academic resources. Customers subscribe to these services and pay a subscription fee, which serves as the main source of revenue.

3. Advertising-Supported Revenue Model:

E-businesses provide free services or information to a specific audience and generate revenue through advertising on their website. Advertisements support the operational costs of the business.

4. Advertising-Subscription Mixed Revenue Model:

E-businesses earn revenue from both subscriptions and advertisements. Subscribers pay a fee for certain services, while other services are offered for free but supported by advertising.

5. Fee-for-Transaction Revenue Model:

E-businesses charge a fee based on the number or size of transactions they process. Examples include online travel agents and stock brokerage firms that facilitate transactions and earn revenue through commissions.

6. Fee-for-Service Revenue Model:

E-businesses charge a fee based on the value of a specific service provided. Professional services offered online by lawyers, doctors, accountants, as well as online entertainment and gaming platforms, follow this revenue model.

Internet Marketing

Internet marketing has rapidly developed, offering marketers new opportunities in the field of e-marketing.

This includes market segmentation, email marketing, banner advertising, promotions, public relations, partnering, customer relationship management, creating brands, affiliate programs, search engines, and global marketing.

Market Segmentation

Market segmentation is the practice of dividing potential customers into specific segments or groups based on certain characteristics. This helps businesses target their advertising messages to specific audience segments. There are several types of market segmentation:

1. Geographic segmentation:

Customers are grouped based on where they live or work, such as urban and rural areas. Different marketing plans may be required for each segment based on their geographic location.

2. Demographic segmentation:

Segmentation is done based on demographic variables like age, gender, income level, and marital status. Different age groups may have different product requirements, so identifying these requirements help in targeting specific groups of customers.

3. Psychographic segmentation:

Customers are grouped based on variables like social class, personality, lifestyle, or approach to life. This type of segmentation helps target customers who have similar lifestyles or preferences. For example, an advertising campaign for a sports car may target customers with a particular lifestyle and a love for thrill.

4. Market Segmentation on the web

Market segmentation is applicable to both e-businesses and physical businesses, as it allows for the creation of unique sales environments for different customer segments. In a physical store, display options, lighting, music, and sales personnel remain consistent for all customer groups. However, the web offers the opportunity to present different store environments online to different customer segments, such as having a specific page for children and a separate page for old people. This allows for separate virtual spaces for different market segments, and some retailers even allow customers to create their own products, a form of one-to-one marketing.

OR

Market segmentation is applicable to e-businesses and physical businesses, offering unique sales environments for different customer segments. Web allows for separate virtual spaces for different market segments, allowing customers to create their own products, similar to one-to-one marketing.

Behavioral segmentation

Another important aspect of market segmentation on web. It involves creating separate experiences for customers based on their online behavior. Three common behavioral modes of customers on the web are:

#Imp

Browsers: Customers who are casually browsing through the site with no immediate intention to make a purchase.

Buyers: Customers who are ready to make a purchase immediately.

Shoppers: Customers who are motivated to make a purchase but require more information before making a decision.

To effectively target customers in different behavioral modes, an e-business site should devise the right combination of marketing strategies to attract and convert visitors into customers.

Choosing a Domain Name

Choosing a suitable domain name is crucial when starting an online business. Since the web has a worldwide reach, it's important to select a domain name that can be easily recognized, remembered, and typed by people from different countries and cultures.

Marketing Research

Marketing research plays a vital role in understanding the strengths and weaknesses of your business and your competitors. It involves conducting interviews, surveys (paper or phone), questionnaires, and analyzing findings from previous investigations. In the case of e-businesses, marketers have the advantage of accessing and analyzing information about the industry, customers, and competitors with just a few clicks. This accessibility to information can be highly beneficial for the success of an e-business.

Web design

E-business success relies heavily on a good web design, as competitors are easily accessible and can quickly switch to a competitor's site.

To maintain a competitive edge,

- Site navigation should be easy, with a map
- FAQs and contact information.
- Multimedia elements should be used with a focus on loading time.
- Privacy policy should outline the intended use of customer personal details.
- The overall website should be attractive and sticky.

Lesson 31

E-MAIL MARKETING

E-mail marketing is a cost-effective way to reach potential customers globally. It allows for personalized communication and can be outsourced if needed. E-mails can improve customer service, provide order updates, and be combined with traditional marketing methods. However, spamming is illegal and should be avoided. Overall, e-mail marketing is a powerful tool for targeting customers and promoting products or services.

1. **Promotions** in e-business are aimed at attracting visitors and encouraging them to make purchases. Here are some popular promotional methods:
2. **Frequent-flyer miles:** Customers earn airline miles for free when they make purchases from the online business up to a specified value.
3. **Point-based rewards:** Customers earn points for performing certain actions and can redeem them for rewards such as t-shirts, mugs, or items with the company's logo.
4. **Discounts:** Advertisements offering discounts in magazines, newspapers, websites, etc., can attract new and repeat customers.
5. **Free trials:** Customers can try out a service or product for free for a limited period of time before making a purchase decision.
6. **Free shipping:** Certain items are shipped to customers without any shipping cost as an incentive.
7. **Coupons:** Online coupons are made available on popular sites, allowing customers to use them for discounted shopping on specific websites.

E-business advertising

Involves disseminating information to effect a buyer-seller transaction, using traditional media like televisions, movies, newspapers, and magazines
Telemarketing and direct marketing personalize advertising, requiring unique, easy-to-remember brands and promoting URLs on direct mails or business cards.

OR

E-business advertising is about sharing information to encourage sales. It's a one-way, paid mass communication by sponsors. Traditional methods include TV, movies, newspapers, and magazines.
Telemarketing and Direct marketing personalize ads for better results. A unique, memorable brand and promoting your website's URL on direct mails or business cards can boost brand awareness.

Banner Advertising

Banner advertising involves placing graphical ads resembling small billboards on popular websites. These ads are mutually beneficial for both the advertiser and the host site. Banners come in different sizes and positions on web pages and use visual techniques to grab attention. They can include links and business logos to increase brand recognition. Marketers should consider ad placement and click-through rates. Banner exchanges and free advertising options are available, and hosting related business banners can indirectly benefit each other. Overall, banner advertising is an effective way to increase brand visibility and attract potential customers online.

Advertising payment modes

Advertising payment modes include monthly charges for online advertising, such as CPM (cost per thousand)

Pay-Per-Performance that includes: #Imp

Pay-per-click – fee according to no. of click to your site

Pay-per-lead – pay for every lead generated from the advertisement

Pay-per-sale – pay the host for every sale resulting from a click through

Exchanging advertising space on your site for advertising space on another's site

The hosting server's log file contains a column for the IP address of the top-referring website, allowing analysis of the number of visitors diverted to the advertising site from the host site.

Web casting#Imp

Web casting refers to internet-based broadcasting of audio and video content, enabling two-way communication between broadcasters and viewers. It is particularly useful for marketers with slow internet access. Video conferencing is an example of web casting, allowing doctors in different countries to exchange ideas in real-time, and benefiting from each other's experiences.

Interactive Advertising

Interactive Advertising combines rich media and traditional forms like print, TV, or radio ads to involve customers in the advertising process and increase brand recognition. For instance, a famous business uses TV commercials to encourage viewers to visit its website, where they can download action pictures and background music. This marketing technique aims to increase brand recognition for customers.

E-business Public Relations

Public Relations (PR) is crucial for keeping customers and employees informed about products, services, and internal and external issues such as any promotional activities, new products, customer reactions.

It can be achieved through various methods, including press releases, speeches, special events, email, chat sessions, and bulletin boards.

Press releases provide important news to the press or media, while video clips of news appearances, speeches, and commercials can also be effective publicity. By utilizing these channels, businesses can effectively reach their target audience and maintain a strong online presence.

Lesson 32

CUSTOMER RELATIONSHIP MANAGEMENT (CRM) #Imp

CRM is the collection of customer service solutions in a company. It provides fast and effective service, including call handling, sales tracking, and transaction support. Tools like log file analysis, cookies, and data mining improve customer service. CRM includes call centers with representatives accessible via phone, email, or chat. Feedback on ad views and clicks can be obtained. CRM aims to establish lasting customer relationships, build loyalty, and trust. Loyalty develops through **five stages** but can decline in the fifth stage, leading to relationship termination. Maintaining strong customer relationships is crucial for long-term success.

These stages are following, #Imp

Awareness: Customers recognize the company or its products but have not yet interacted with it.

Exploration: Customers have gained more knowledge about the company or its products, possibly through visiting the website or exchanging information.

Familiarity: Customers have engaged in multiple transactions with the company and are familiar with its policies and offerings.

Commitment: Customers have developed a strong sense of loyalty and preference for the company's products or brand. They may advocate for the company and its offerings.

Separation: Conditions change, leading to customer dissatisfaction or high maintenance costs. The relationship enters the separation stage.

Life Cycle Segmentation:

The customer life cycle consists of the five stages mentioned earlier, and companies can use this information to segment their customers and develop better relationships. By understanding where customers are in the life cycle, companies can customize their products or services accordingly.

B2B Marketing on the Web:

Effective CRM requires integration between different steps in a customer transaction, including selling, buying, marketing, front-end, and back-end operations. In B2B marketing, there is no direct contact with end users, unlike B2C. This lack of direct contact affects the feedback and response received by an e-business. Marketing plans differ between B2B and B2C due to this distinction.

Search Engines:

Search engines are programs that scan websites and create lists of relevant sites based on keywords or ranking criteria. They help users find specific information from the vast amount available on the internet. Popular search engines like Google, Altavista, and Yahoo provide search engine services. As a marketer, it's important to register your e-commerce website with search engines to ensure its visibility in search results.

Lesson 33

META INFORMATION

#Imp

Meta tags contain key information about a web page used by search engines to locate and rank websites. By submitting Meta information to search engines, websites can improve their visibility. Different search engines have their own ranking criteria, and sites with matching keywords and higher frequency in Meta information are ranked higher. Misusing competitors' Meta information to improve ranking is unethical and can be considered passing-off, an offense in many countries.

Partnerships

Partnerships involve forming strategic unions with another company or business for mutual benefit or profit sharing. By providing complementary services and products, partners can benefit each other. This competitive advantage allows customers to be a link away from buying complementary products or services. Partners can exchange technical research and customer information, improve management or operations, and outsource jobs.

Affiliate Programs

Affiliate programs involve a commission-based agreement between two parties. They are used to drive customer actions, such as clicks or purchases. It is a time-limited arrangement where one party pays the other for promoting their products or services. Banner advertising is a common example of an affiliate program.

Branding

Branding is the process of creating and maintaining a distinct identity for a company or its products/services. It involves establishing a recognizable name, symbol, or image that represents the company's reputation and creates an emotional connection with customers.

Elements of Branding #Imp

Researchers have identified three elements of branding, that is,

Differentiation

Relevance

Perceived Value

Product Differentiation: Creating a unique feature or characteristic that sets a product apart from competitors.

Relevance: The degree to which a product or service meets the needs and interests of potential customers.

Perceived Value: The subjective worth or benefits that customers attribute to a product or service based on their perceptions and experiences.

Emotional branding VS rational branding#Imp

Emotional Branding: Creating a brand image or connection with customers based on emotions and feelings. It is commonly used in traditional media where the audience is more passive.

Rational Branding: Focusing on logical and practical aspects of a brand to appeal to customers. It is often used in online marketing where customers have more control and can easily navigate away from emotional appeals.

Managing existing brands online is simpler and less expensive than creating a new brand. Promoting a brand on the web involves including the website URL on product packaging, advertisements, and combining it with the company logo on promotional materials.

Global Marketing

To effectively market your business globally, consider the language and currency of your web design, as well as the infrastructure needed for e-business operations. Regular currency exchange rate displays and tools for translating emails and websites can help reach diverse audiences. Additionally, choose a payment system that is compatible with the country's business environment, as SET payment mechanisms may not be suitable in some third-world countries.

Lesson 34

DATA MINING

#Imp

Data mining is the process of discovering patterns in large amounts of data, enabling businesses to identify customer behavior and target them with personalized messages. For example, a shopping store can use data mining techniques to identify demographic profiles and preferences for computer-related items. This information can be used by marketers to focus their marketing efforts on these individuals, saving time, energy, and mailing costs.

Data warehouse#Imp

A data warehouse is a repository for long-term storage of data from multiple sources, organized for decision-making management. It involves cleaning, transforming, integrating, and loading data from various sources, allowing clients to access it for data mining and pattern evaluation.

Knowledge discovery

Knowledge discovery involves data cleaning, integration, selection, transformation, mining, pattern evaluation, and presentation of information.

Data mining is a crucial step in the knowledge discovery process, requiring data cleaning, transformation, selection, and integration. Data cleaning involves replacing erroneous values with correct ones, while selection involves discarding or deleting erroneous rows. Data transformation converts data from various sources into a consistent format, and integration assembles data into a warehouse.

Types of Data Mining #Imp

There are four main types of data mining as follows:

- 1. Classification**
- 2. Association**
- 3. Characterization**
- 4. Clustering**

Classification

Classification is a predictive modeling technique that uses algorithms like ID3 and Bayesian classification to assign labels to samples. By analyzing data on factors like sex, education, and income, the model can accurately predict the likelihood of belonging to different classes, such as males and females. The model achieves a high probability of correctly classifying males, which saves time, energy, and costs in targeted outreach. The efficiency of the model is evaluated by comparing actual and theoretical values, and in this case, the model shows good efficiency at 75.75%, indicating its reliability for future use.

Association

Association analysis is a technique used for identifying common attribute-value conditions in data, such as market basket analysis. It helps identify strong bondages or affinity between items, such as strong purchases of one item. The Apriori algorithm is commonly used for association mining.

Lesson 35

CONFIDENCE AND SUPPORT

In association mining, support and confidence are important measures. Confidence measures how often a relationship holds

true, **Support** measures the overall occurrence of two items together. By setting desired percentages for confidence and support in an association mining tool, marketers can identify items with significant associations. They can then use this information to improve customer service, design promotional schemes, and make informed marketing strategy decisions.

Characterization

Characterization is a technique used to discover generalized concepts by examining data. It helps in understanding the overall behavior of the data by providing concise and generalized answers. **For example**, it can tell us how many students from a specific country are studying a particular field of education. By applying this technique, we can extract meaningful insights from the data.

The two algorithms used in characterization are Version Space Search and Attribute-Oriented Induction

Clustering#Imp

Clustering is a technique that groups similar data objects together, while keeping dissimilar objects in separate clusters. It is useful for customer grouping, email categorization, and web usage analysis. The K-means algorithm is commonly used for clustering.

Online Analytical Processing (OLAP) #Imp

OLAP (Online Analytical Processing) uses domain knowledge to present data at different abstraction levels. It helps decision-

makers and managers view data conveniently. OLAP displays data in a data cube format, showing information across dimensions. Processes like drill-down, roll-up, slice, and dice are used to explore and summarize data in an OLAP cube.

SAS (Enterprise Miner) and DB miner: #Imp

SAS (Enterprise Miner) and DB miner are popular tools for data mining and OLAP. Characterization can be used with any data type, while OLAP primarily works with numeric data.

Lesson 36

ELECTRONIC DATA INTERCHANGE (EDI)

#Imp

Electronic Data Interchange (EDI) is a technology used by organizations to exchange business documents electronically. It eliminates paper-based transactions and human intervention, allowing for quick and automated exchanges. EDI documents follow standardized formats like ANSI X12 and EDIFACT. The data is encoded and structured, enabling seamless communication between trading partners. EDI improves efficiency and accuracy in transactions, benefiting companies and their suppliers.

Example (Just read this example and understands about EDI)

Company A and Company B, involved in regular transactions. Company A wants to place an order for a specific product with Company B. Instead of using traditional paper-based methods, they decide to use EDI for their transactions.

Using EDI, Company A electronically sends a purchase order to Company B in a predefined format. The purchase order includes information such as the product details, quantity, and any specific instructions. This electronic document is sent through specialized EDI software or systems.

Company B receives the electronic purchase order in seconds and their system automatically processes it. They can quickly view the details of the order, check their inventory, and generate an electronic confirmation or acknowledgment that the order has been received and will be processed.

The entire transaction occurs without any paper documents or manual intervention. The information exchange between Company A and Company B happens seamlessly, saving time, reducing errors, and improving efficiency.

Value Added Network (VAN) #Imp

Value Added Networks (VANs) are third-party networks that facilitate authorized transactions using EDI. They provide services to execute transactions between trading partners. VANs maintain postboxes for outgoing messages and mailboxes for

incoming messages. Users can send messages to their trading partners, which are temporarily stored in their postboxes. The VAN system inspects the postboxes, repackages the messages, and moves them to the recipients' mailboxes.

Advantages of VAN

This enables secure and efficient communication between trading partners, offering advantages such as time independence and protocol compatibility.

1. **Time Independence:** Users can send and receive interchanges or messages at their convenience, without the need for simultaneous connectivity. This allows for flexibility in transaction timing.

2. **Protocol Independence:** VANs re-envelop interchanges with the appropriate transmission protocol when they are retrieved from the postbox. This ensures protocol compatibility between the sender and the recipient, even if their systems use different protocols.

Internet-Based EDI#Imp

Internet-based EDI utilizes the internet for EDI transactions, replacing the need for a VAN. It can involve using internet email, creating extranets for web form data entry, or utilizing

web-based EDI hosting services. Further efforts are needed to promote and popularize internet-based EDI.

Benefits of EDI include:

Shortened Ordering Time: EDI automates the order generation process, reducing the time required for placing orders.

Cost Cutting: EDI transactions are more cost-effective as they eliminate expenses associated with paper-based transactions and the need for additional staff. The primary cost lies in the implementation of EDI software.

Elimination of Errors: With automated message generation, the chances of human-induced errors, such as typing mistakes, are significantly reduced.

Fast Response: EDI messages can be electronically read and processed by receivers using EDI software. This enables quick fulfillment and implementation of orders, particularly for suppliers of raw materials.

Accurate Invoicing: Invoices or payment requests can be generated using the standardized EDI format, ensuring accuracy and reducing discrepancies compared to paper invoices.

EDI Payment: EDI standard documents can be used to electronically provide financial information for payment purposes.

Electronic Banking

Electronic banking, also known as cyber banking, virtual banking, home banking and online banking, includes various banking activities conducted from home, business, or on the road, instead of at a physical bank location.

Advantages of e-banking

E-banking offers numerous benefits, including accessing current account balances, accessing credit card statements, paying utility bills, downloading account information, transferring money, sending emails, managing schedules, handling finances remotely, and applying for loans online.

It offers an affordable alternative to branch banking and enables remote banking.

Lesson 37

PERSONAL FINANCE ONLINE

In personal finance, software packages such as Quicken, MS Money, and Money 2003 are used to manage financial matters, including tax calculations and budgeting. These software packages can import online data automatically and help with organizing transactions, calculating taxes, and preparing budgets.

Value Chain#Imp

Value chain analysis, introduced by Michael Porter, breaks down a business into different value-adding activities. A strategic business unit combines a specific product, distribution channel, and customer type. The value chain consists of primary activities and support activities.

Primary activities include:

- Identifying customers through market research and surveys
- Designing products from concept to manufacturing
- Purchasing materials and supplies, including vendor selection and quality monitoring
- Manufacturing products or creating services through fabrication, assembly, and packaging
- Marketing and selling activities such as advertising, promotions, and pricing
- Delivering products through warehousing, shipment, and timely delivery
- Providing after-sales service and support, including installation, repairs, and warranties

Support activities provide infrastructure for the primary activities

- Finance and administration activities, including accounting and complying with regulations

- Human resources management, such as recruitment, compensation, and benefits
- Technology development to improve products/services and enhance process efficiency

The importance of each primary activity depends on the nature of the product/service and the target customers. Support activities contribute to the overall functioning of the primary activities.

Industry value chains

Industry value chains analyze the activities of different business units within an industry. By understanding these value chains, businesses can identify opportunities for cost reduction, product improvement, and new market reach.

Electronic commerce (EC) can contribute to value chains by reducing costs, improving product quality, reaching new customers and suppliers, and creating new sales channels.

For example, a software developer can eliminate software retailers by directly distributing updates to customers online. This enhances efficiency and revenue. Overall, analyzing value chains and utilizing EC can lead to strategic advantages in the industry.

SWOT (strengths, weaknesses, opportunities and threats) analysis

SWOT analysis is a strategic tool used to assess the strengths and weaknesses of a business unit and identify opportunities and threats in its operating environment.

Example of Dell

In the case of Dell, the analysis revealed strengths in direct sales, efficient design, and customization, while weaknesses included limited relationships with local dealers and potential financial liabilities. Opportunities were found in industry trends, new markets, and emerging technologies, while threats included competition from established brands and potential changes in the business environment. By leveraging its strengths and addressing weaknesses, Dell developed a successful strategy focused on customized computers, utilizing direct sales and internet marketing.

Lesson 38

SUPPLY CHAIN

The supply chain encompasses the flow and transformation of goods from raw materials to the end user. It consists of three main parts: upstream activities, internal activities, and downstream activities. **#Imp**

Upstream activities involve the sourcing of materials or services from suppliers.

Internal activities refer to the manufacturing and packaging processes within the business.

Downstream activities involve the distribution and sale of goods to distributors and customers.

By understanding and managing the entire supply chain, businesses can ensure a smooth flow of goods, optimize efficiency, and deliver products to customers in a timely manner.

Supply chain management **#Imp**

Supply chain management involves actively working with suppliers to improve products and processes, and it focuses on adding value to the end customer. It includes establishing long-

term relationships with capable suppliers, optimizing various supply chain activities, and considering sustainability and ethical factors. Effective supply chain management leads to operational excellence, customer satisfaction, and competitive advantage.

Internet technologies and supply chain#Imp

Internet technologies play a crucial role in supply chain management by facilitating quick and effective communication.

They enable suppliers to share information about customer demand and product changes, exchange drawings and specifications, increase transaction processing speed, reduce transaction costs, and minimize errors in data entry. While there may be some associated costs, the advantages of using internet technology in the supply chain outweigh the drawbacks. Supply chain management software further enhances the ability to manage internal processes and predict production needs within the supply chain.

Examples of technology use in supply chain

Technology use in supply chain management includes leveraging information systems and internet technology to improve coordination, reduce production delays, shorten delivery times, enhance customer service, reduce inventory costs, and increase production efficiency. Examples such as the largest producer of commercial aircraft and a famous computer selling brand demonstrate the benefits of technology in enabling information sharing, collaboration, and better planning among

supply chain members. Overall, technology plays a crucial role in optimizing supply chain operations and improving overall performance.

Supply chain and ultimate consumer orientation

Supply chains should have an ultimate consumer orientation, focusing on meeting the needs of the end consumer. A tire production company, for example, shifted its focus to tire dealers by providing them with an extranet to access tire specifications, inventory status, and promotional information directly. This approach improved service to dealers, reduced costs, and increased customer recommendations, ultimately benefiting the end consumer.

Competitive Strategy

Competitive strategy is the approach a business takes to gain a competitive advantage over its rivals. The three basic strategies for competitive advantage are cost leadership, differentiation, and focus. **#Imp**

Cost leadership involves offering goods or services at a lower price than competitors, attracting more customers through affordability.

Differentiation focuses on offering a product or service with unique qualities or features that make it more appealing than competitors, even if the competitor's price is slightly lower.

Focus strategy concentrates on a specific aspect of the market, such as a particular market segment, geographic area, or product type, in order to achieve higher profitability and a stronger distribution channel.

Role of e-commerce in Competitive Strategy

By applying EC following major benefits can be derived#Imp

E-commerce (EC) supports competitive strategies by reducing administration and transaction costs; improving logistics supply chain, enabling customization, and differentiating products or services in terms of service quality. EC helps achieve cost leadership, differentiation, and focus strategies by reducing costs, enhancing efficiency, and providing personalized customer experiences.

Lesson 39

PORTER'S MODEL OF COMPETITIVE RIVALRY

Porter's Five Forces model helps firms identify threats to their competitive position, and IT and e-commerce can be utilized to protect or enhance that position by addressing the forces of new entrants, substitutes, buyer power, supplier power, and competition.

Five forces analysis assesses the threats to a firm's

competitive position: #Imp points

- **Threat of new entrants:**

IT/EC can act as a barrier for new entrants, but advancements in technology also provide opportunities for new entrants without heavy infrastructure investment.

- **Threat of substitution:**

New products or services can substitute existing ones, such as e-commerce substituting physical music shops or online banking substituting traditional banking.

- **Bargaining power of buyers:**

IT/EC can help reduce production costs and afford lower prices, improving bargaining power if there are many competitors and/or surplus supply.

- **Bargaining power of suppliers:**

Electronic enablement and plentiful supply can reduce supplier bargaining power.

- **Competition between existing players:**

EC can reduce administration/transaction costs, improve supply chain efficiency, and enhance product quality and customer service, making a significant difference in competition.

These five forces analysis helps determine the attractiveness of an industry and whether a business should enter it or not.

Strategic Planning Cycle#Imp

E-business competitive strategy is normally formed and implemented according to a planning cycle which is called strategic planning cycle.

There are four stages in this planning cycle:

Industry and competitive analysis help identify factors for the success of an EC project or business.

Strategy formulation involves developing a competitive strategy based on internal and external analysis.

Implementation includes creating a plan and taking steps to put the strategy into action.

Performance Assessment or Strategy Reassessment involves monitoring results, making adjustments, and learning from feedback. The strategic planning cycle ensures alignment with objectives and adaptability to the changing environment.

Lesson 40

BARRIERS TO INTERNATIONAL E-COMMERCE

There are several barriers to international e-commerce, including lack of trust, lack of infrastructure, language, and culture. **#Imp Points**

Lack of trust arises from the anonymity of online businesses, and establishing credibility through web design and customer-focused practices is crucial.

Language barriers can be overcome by providing local language versions of the website using translation tools.

Cultural considerations are important, such as choosing appropriate names, icons, and colors that are acceptable and relevant in different cultures. Cultural and legal restrictions, such as government censorship and compliance with local laws, also impact international e-commerce operations.

Infrastructure issues, such as limited internet infrastructure and high costs, can hinder international transactions. Coordinated efforts with customs, freight carriers, and government officials may be required to navigate complex regulations. Overcoming these barriers requires addressing trust, language, cultural, and infrastructure challenges to successfully engage in international e-commerce.

Electronic Transactions Ordinance, 2002 (ETO) #Imp

The Electronic Transactions Ordinance, 2002 (ETO) is a law in Pakistan that provides legal recognition to documents in electronic form and electronic signatures. It defines various terms related to electronic transactions and establishes the framework for their validity and authenticity. Some key terms defined in the ETO include:

- **Certificate:** A certificate issued by a Certification Service Provider to confirm the authenticity or integrity of electronic documents or electronic signatures.
- **Cryptography services:** Services related to transforming the contents of electronic documents into a form that cannot be understood by unauthorized individuals.
- **Accredited Certification Service Provider:** A Certification Service Provider that is accredited under the ETO to issue certificates for the use of its cryptography services.
- **Certification Practice Statement:** A statement prepared by a certification service provider that outlines the practices it employs for issuing certificates.
- **Originator:** The person who generates or sends an electronic document before it is received or stored.
- **Addressee:** The person intended to receive the electronic communication.
- **Information system:** An electronic system used for creating, generating, sending, receiving, storing, or processing information.
- **Electronic Signature:** Letters, numbers, symbols, images, or characters in electronic form that are applied to or associated with an electronic document to authenticate or approve it.
- **Authenticity:** The identification and attribution of an electronic document or electronic signature to a specific person or information system.
- **Integrity:** The assurance that an electronic document or electronic signature has not been tampered with, altered, or modified since a specific point in time.

Appropriate authority' means

The term "Appropriate authority" refers to the governing body or entity responsible for overseeing specific functions or matters related to electronic transactions.

OR

'Appropriate authority' means the relevant entity responsible for making decisions:

- For Federal Legislative List matters: Federal Legislature or Federal Government.
- For Concurrent Legislative List matters with a Federal law: Federal Legislature or Federal Government.
- For other Concurrent Legislative List matters: Respective Provincial Legislature or Provincial Government.
- For functions handled by a statutory body: That specific statutory body.
- For matters regulated by the Supreme Court or High Courts: Supreme Court or High Court, as applicable.

Section 3 of the ETO provides:

Section 3 of the ETO states that no document, record, information, communication, or transaction shall be denied legal recognition, admissibility, effect, validity, proof, or enforceability solely because it is in electronic form and does not have a witness's attestation.

Section 4 of the ETO provides:

Section 4 of the ETO states that if any law requires a document, record, information, communication, or transaction to be in written form, that requirement is considered fulfilled if the document is in electronic form and accessible for subsequent reference.

Sections 3 and 4 together mean that the legal requirement for a document to be in writing can be satisfied by having it in electronic form, as long as it is accessible and usable for reference.

Lesson 41

ELECTRONIC TRANSACTIONS ORDINANCE, 2002 (ETO) (CONTINUED....)

In Pakistan, the term 'Appropriate authority' generally refers to legislative assemblies (national and provincial), the federal and provincial governments, the Supreme Court, High Courts, and statutory bodies related to governmental functions. Statutory bodies are established by laws. For example, the Lahore Development Authority (LDA) is one such body created under a provincial statute.

The Constitution of Pakistan is the highest law, requiring all other laws to adhere to its terms. It has Federal and Concurrent legislative lists, indicating subjects for federal and joint federal-provincial legislation, respectively.

Section 6 of the ETO

Section 6 of the Electronic Transactions Ordinance (ETO) sets conditions for validly retaining electronic documents. These include accessibility for reference, accurate representation of original content and form, and the ability to identify origin, destination, date, and time of generation. This ensures electronic documents are usable, authentic, and traceable.

Section 7 of the ETO

Section 7 of the ETO recognizes electronic and advanced electronic signatures as valid for meeting signature requirements. Advanced electronic signatures are presumed authentic. The difference lies in involving an accredited certification service for advanced signatures. Traditional legal documents require paper and stamp duties under the

Stamp Act, 1899. The Qanoon-e-Shahadat Order, 1984, often needs male witnesses for documents. Some documents can be notarized by a notary public to validate copies.

Sections 10 and 11 of the ETO

Sections 10 and 11 of the ETO waive stamp duty and attestation/notarization requirements for two years or until electronic payment methods and proper attestation/notarization measures are established by provincial governments, whichever is later. This encourages electronic document usage and simplifies their legal processes.

Section 13 of the ETO

Section 13 of the ETO defines who is considered to have sent an electronic communication:

1. An electronic communication is deemed from the originator if sent by:
 - The originator themselves,
 - Someone authorized to act for the originator regarding that communication, or
 - An automated system set up by or for the originator.
2. The recipient should consider the electronic communication as from the originator and can act on this assumption if:
 - There's no reason to doubt its authenticity, or
 - There are no circumstances indicating it's not authentic that the recipient should have known with reasonable care.

Lesson 42

ELECTRONIC TRANSACTIONS ORDINANCE, 2002 (ETO) (CONTINUED....)

Section 14 is the relevant provision in this behalf:

Section 14 outlines the rules for electronic communications and acknowledgments. If the sender states that an acknowledgment is required, the communication is treated as unsent until the acknowledgment is received. If no specific acknowledgment method is agreed upon, the recipient can acknowledge through any form of communication or conduct indicating receipt.

Section 15 of the ETO

Section 15 of the ETO outlines rules for the time and place of dispatch and receipt of electronic communications. It establishes guidelines for determining when an electronic communication is considered sent and received, and how the place of dispatch and receipt is determined, which is important for territorial jurisdiction and compliance with time limits for legal actions.

Section 16

Section 16 of the law states that individuals cannot demand that authorities handle documents in electronic form. However, when an appropriate authority permits electronic methods for documents, permits, payments, etc., it can specify the details, such as formats and security procedures. This provision supports e-government initiatives.

Section 17 and 18

Section 17 permits certification service providers to offer services without accreditation.

Section 18 mandates the creation of a Certification Council by the Federal Government, composed of five members. This council will oversee accreditation and perform various functions as outlined in Section 21.

“19. of the five members of the Certification Council:

&

“20. the funds of the Certification Council shall comprise of:

Section 19 outlines the qualifications for the five members of the Certification Council, covering fields like telecommunications, IT, administration, and law.

Section 20 details the funding sources for the Certification Council, including government grants, fees for accreditation, and fines.

“21

Section 21 outlines the Certification Council's functions, including granting accreditations, monitoring compliance, and managing the repository.

Section 23 in the ETO

Section 23 mandates the establishment and management of the repository

containing accreditation and certificate information, which is open to public inspection.

Lesson 43

ELECTRONIC TRANSACTIONS ORDINANCE, 2002 (ETO) (CONTINUED....)

Section 24 of the ETO

Section 24 of the ETO allows the Certification Council to establish regulations for accrediting certification service providers. These regulations cover criteria, procedures, terms, fees, and a transparent process with a right to a fair hearing for providers seeking accreditation.

Under Section 25

Section 25 requires certification service providers to create a Certification Practice Statement (CPS) according to regulations. CPS includes policies on notifying affected parties, accuracy, and more. CPS is submitted with accreditation applications, available for public inspection, and can be changed with Council approval. Providers must comply and ensure accuracy during certificate validity. Council can revoke accreditation for non-compliance after a fair hearing. All matters should be decided promptly with reasons. The Council can appoint personnel and establish regional offices for efficient functioning.

Section 31 of the ETO

Section 31 of the ETO excludes certain documents from its application:

1. Negotiable instruments (promissory notes, bills of exchange, checks).
2. Power of attorney documents.
3. Trusts (excluding certain types).
4. Wills and testamentary dispositions.
5. Contracts for sale or conveyance of immovable property.

The Federal Government can choose to apply the ETO to these documents partially or fully through a notification.

Section 32 of the ETO

Sections 32 and 33 are reproduced as under in this behalf

Section 32 of the ETO grants Pakistani courts authority over matters related to the ordinance even if they occur outside Pakistan but affect persons, information systems, or events within Pakistan's jurisdiction. Section 33 asserts the ETO's primacy over conflicting laws.

Sections 34 to 37 of the ETO

Sections 34 to 37 of the ETO outline four types of offenses:

1. Providing false information to a certification service provider.
2. Directors/officers issuing false certificates or not revoking misleading ones.
3. Unauthorized access to information systems.
4. Unauthorized alteration or impairment of information systems.

Penalties involve imprisonment, fines, or both, depending on the offense.

“35.

Section 35 of the ETO addresses directors and officers of certification service providers who commit offenses:

1. Issuing false/misleading certificates.
2. Failing to revoke/suspend false/misleading certificates.
3. Issuing certificates with suspended/revoked accreditation.

Penalties include imprisonment up to seven years, a fine up to ten million rupees, or both. Compensation may be required for damage caused by their actions.

Section 36 of the ETO addresses unauthorized access to information systems, punishable with imprisonment up to seven years, a fine up to one million rupees, or both.

Section 37 Involves unauthorized actions to alter, delete, impair, or hinder access to information systems, also punishable with imprisonment up to seven years, a fine up to one million rupees, or both.

Lesson 44

GLOBAL LEGAL ISSUES OF E-COMMERCE

The ETO assigns jurisdiction to session courts in Pakistan. Pakistan's court hierarchy encompasses civil and criminal courts, with civil law pertaining to private rights and criminal law aiming to punish wrongdoers. Civil courts handle civil wrongs, while magistrates handle criminal cases. Appeals can be made to district and session courts, followed by high and Supreme Courts. The High Court has extraordinary writ jurisdiction under Article 199 of the Constitution.

Globally, many countries recognize electronic documents and signatures following UNCITRAL's e-commerce model law from 1996. The ETO has shortcomings like unclear electronic signature distinctions and broadly-worded privacy and damage sections. It lacks international e-commerce considerations and a clear differentiation between accredited and non-accredited certification providers. Rules and regulations under the ETO are still pending.

Key global e-commerce legal issues include jurisdiction, online contracts, copyright, domain and trademark disputes, online defamation, privacy, internet taxation, and cybercrimes.

OR

The ETO establishes jurisdiction with session courts in Pakistan. The court hierarchy involves civil and criminal courts, with appeals moving through district, session, high, and Supreme Courts. Global e-commerce legal issues include jurisdiction, online contracts, copyright, domain disputes, defamation, privacy, internet taxation, and cybercrimes. The ETO has some shortcomings, such as unclear electronic signature distinctions and missing international aspects of e-commerce.

Territorial Jurisdiction

Territorial jurisdiction in legal matters involves a court's authority based on location. In cyberspace, where borders don't apply, disputes can arise. Plaintiffs can choose the most favorable jurisdiction, a practice called forum shopping. Courts consider interactivity, commercial nature, and effects of information exchange to establish jurisdiction. This is complex when dealing with disputes spanning different countries and legal systems.

Online contracts

Online contracts require offer, acceptance, and consideration, just like physical contracts. Clicking "I accept" on a webpage can create a valid contract. Contracts can be oral, written, or through actions. Web ads usually invite offers. Digital signatures help verify identity and authority. Companies must protect passwords to prevent unauthorized acceptances. Negligence might make a company liable for breaches and compensation.

Copyright in cyberspace

Copyright in cyberspace is complex due to the internet's nature. Copyright grants exclusive rights to creators of works. Ideas aren't copyrighted, only their expression. Web pages can be copyrighted, but copying challenges arise online. 'Fair use' exceptions apply, allowing limited use for criticism, teaching, etc. 'Fair use' helps address internet copyright issues. Hyperlinks may imply permission for downloading or copying content.

Generally, the protection under 'fair use' may be sought on the following basis,

Fair use protection is based on factors like educational use, factual content, limited copying, and minimal impact on market value. Proper citation is crucial to avoid plagiarism. The 'Napster case' highlighted copyright issues in online music sharing, resulting in legal action and shutdown. WIPO proposed treaties for global copyright protection in response to digital

advancements.

Lesson 45

GLOBAL LEGAL ISSUES OF E-COMMERCE

Patent infringement

A patent provides exclusive rights to inventions. It must be unique and useful. Software patents are complex and copyright registration is often chosen. 'Business process patents' protect unique business methods. They can raise concerns about monopolies. Legal issues around these patents are still being debated.

Trade marks associate businesses with products/services. Domain names are user-friendly web addresses. Conflicts arise between them due to uniqueness.

Cyber squatting involves registering trademark-related domains for ransom.

Concurrent use occurs when legitimate parties can't share a domain due to uniqueness.

Parasites

Parasite domain names mimic famous ones for advantage. Mistypes lead to unintended sites. ICAAN uses UDRP for disputes, allowing trademark holders to retrieve domains via arbitration. WIPO is one such arbitration provider.

Online Defamation

Online defamation harms reputation. False statements about a person or product are defamatory. Anonymous online nature complicates cases. Justifiable criticism vs. defamation is blurry. Web sites should avoid negative statements, altering images, or defaming competitors. ISPs' liability linked to editing control: liable as

a publisher, no control means any liability.

Online Privacy

Online privacy concerns evolve with internet growth. Websites can gather visitor data, raising privacy issues. Different global privacy expectations exist. Countries like Canada, EU have privacy laws, like PIPEDA and EU directive. US lack strict regulations, companies form own policies: opt-out (assume consent) and opt-in (explicit consent) approaches. Principles for privacy laws: improve service, limit data sharing, inform customers, allow data deletion.

Internet taxation affects e-businesses, subject to income tax, sales tax, and property tax. Nexus determines tax liability, especially for cross-border businesses. Double taxation is avoided through credit/refund mechanisms.

Cyber crimes like fraud, hate speech, cyber-stalking, and more emerge due to internet use. Global efforts to draft laws or amend existing ones to address cyber crimes are underway, with territorial jurisdiction challenges.

Some important Key Words regarding IT430 (Final term)

1. E-Commerce: Buying and selling goods/services online.
2. Electronic Commerce: Same as E-Commerce.
3. Online Shopping: Buying products on the internet.
4. Cyber Wallet: Digital wallet for storing and making payments.
5. SSL Handshake: Process to establish secure communication.
6. Encryption: Scrambling data for security.
7. Decryption: Unscrambling encrypted data.
8. Digital Certificates: Verifies identity in online transactions.
9. Payment Gateways: Facilitate online payment processing.
10. Virtual PIN Payment System: Early internet credit card-based payment system.

11. First Virtual Holdings: Company that launched Virtual PIN system.
12. Automated Clearing House (ACH): Centralized system for clearing payments.
13. Client Wallet: User's digital wallet for transactions.
14. Merchant Software: Software for online merchants to manage sales.
15. Public Key Encryption: Asymmetric encryption using public keys.
16. Private Key: Key for decrypting messages in asymmetric encryption.
17. Symmetric Key: Shared key for encrypting and decrypting data.
18. Cipher Suite: Combination of encryption algorithms and methods.
19. SSL/TLS Protocols: Secure communication protocols.
20. Digital Signature: Authenticates digital documents.
21. HTTPS: Secure version of HTTP for encrypted communication.
22. Data Privacy: Protection of personal information.
23. Data Security: Measures to safeguard data.
24. Data Integrity: Ensuring data remains unchanged.
25. Authentication Methods: Methods to verify user identity.
26. Secure Sockets Layer (SSL): Protocols for securing data transmission.
27. Transport Layer Security (TLS): Successor to SSL for secure communication.
28. Hash Function: Converts data into fixed-length hash values.
29. Online Transactions: Financial transactions conducted over the internet.

Key points from 41 to 45

Lesson 41+42: Electronic Transactions Ordinance, 2002 (ETO)

- The term 'Appropriate authority' in Pakistan refers to legislative bodies, governments, courts, and statutory bodies related to government functions.
- Pakistan's Constitution is the highest law, with Federal and Concurrent legislative lists.

- Section 6 of ETO sets conditions for validly retaining electronic documents for accessibility, representation, and traceability.
- Section 7 recognizes electronic and advanced electronic signatures, differing in involving accredited certification services.
- Sections 10 and 11 waive stamp duty and attestation/notarization requirements for electronic documents.
- Section 13 defines electronic communication originator and recipient assumptions.

Lesson 43: Electronic Transactions Ordinance, 2002 (ETO) (Continued)

- Sections 14, 15, and 16 outline rules for electronic communications, acknowledgments, and use of electronic methods.
- Sections 17 and 18 relate to certification service providers' services and the creation of the Certification Council.
- Section 19 lists qualifications for Certification Council members.
- Section 20 details funding sources for the Certification Council.
- Section 21 outlines the Certification Council's functions.
- Section 23 mandates the establishment and management of the repository containing accreditation and certificate information.

Lesson 44: Global Legal Issues of E-Commerce

- Territorial jurisdiction in cyberspace is complex due to the absence of physical borders.
- Online contracts require offer, acceptance, and consideration, and digital signatures help verify identity.
- Copyright in cyberspace involves exclusive rights for creators and 'fair use' exceptions for limited purposes.
- Patent infringement and trademark disputes are significant issues in e-commerce.

- Cyber squatting, concurrent use and parasite domain names lead to domain conflicts.
- Online defamation involves false statements harming reputation, with challenges in distinguishing criticism.
- Online privacy concerns vary globally, with different privacy laws and approaches.
- Internet taxation affects e-businesses with income, sales, and property taxes.
- Cybercrimes like fraud hate speech and cyber-stalking pose challenges and require legal responses.

Lesson 45: Global Legal Issues of E-Commerce (Continued)

- Patent infringement involves protecting inventions, and trademarks associate with products/services.
- Domain name conflicts arise, including cyber squatting and concurrent use.
- Parasite domain names mimic famous ones, leading to confusion.
- Online defamation involves false statements and challenges in the online environment.
- Online privacy concerns evolve with internet growth and different global expectations.
- Internet taxation affects e-businesses subject to various taxes.
- Cybercrimes like fraud and cyber-stalking pose challenges and require global legal efforts.

Feel free to ask me about any queries

Remember me in your prayers

☆☆☆☆☆ Wish you all the best ☆☆☆☆☆